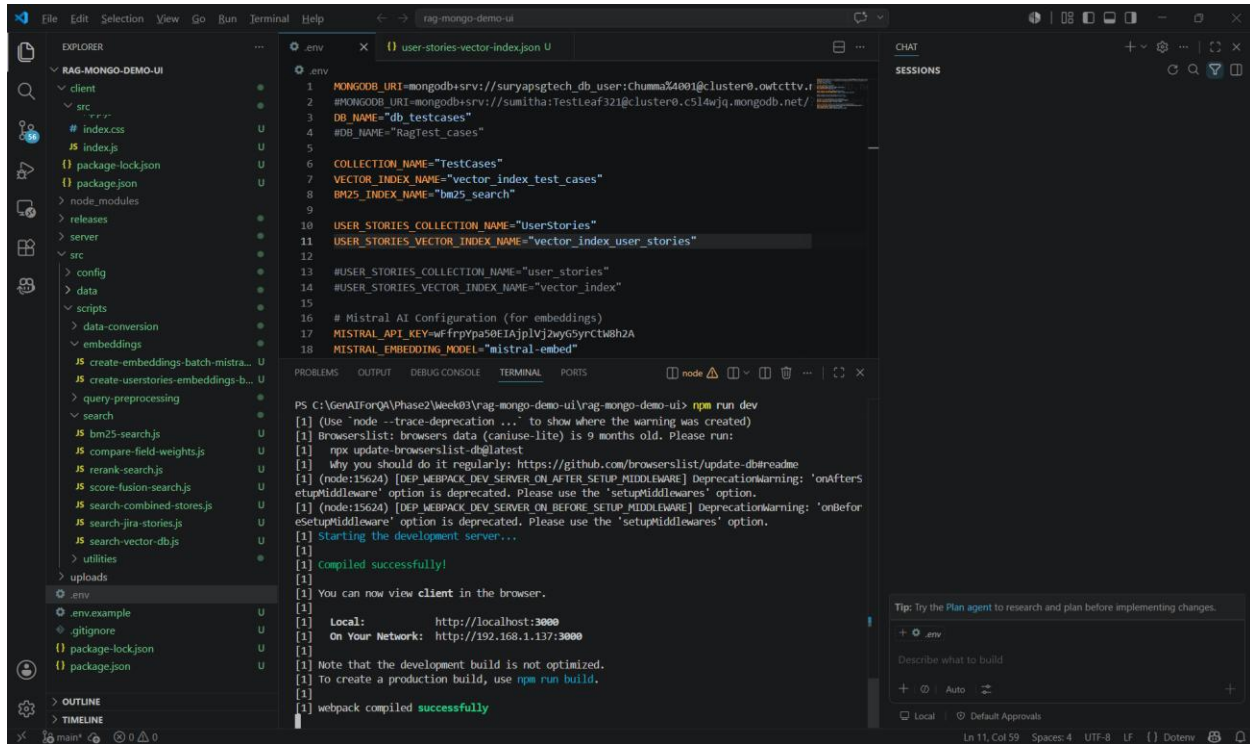


RAG Ingestion – Frontend

Start the server - npm run dev



The screenshot shows a Visual Studio Code editor with the 'rag-mongo-demo-ui' project open. The Explorer panel on the left shows the project structure, including 'client', 'src', 'package-lock.json', 'package.json', 'node_modules', 'releases', 'server', 'config', 'data', 'scripts', 'data-conversion', 'embeddings', 'create-embeddings-batch-mistra...', 'create-userstories-embeddings-b...', 'query-preprocessing', 'search', 'bm25-search.js', 'compare-field-weights.js', 'rerank-search.js', 'score-fusion-search.js', 'search-combined-stores.js', 'search-jira-stories.js', 'search-vector-db.js', 'utilities', 'uploads', '.env', '.env.example', '.gitignore', 'package-lock.json', and 'package.json'. The main editor area shows the '.env' file with the following content:

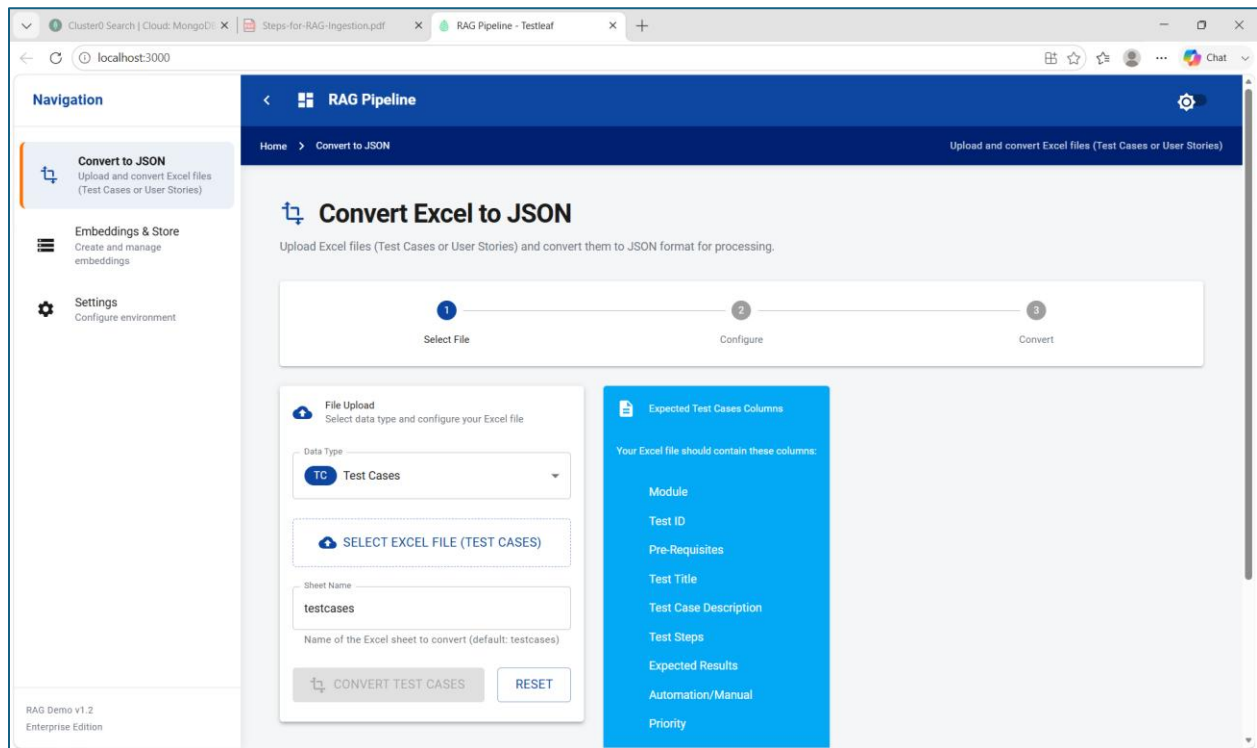
```
1 MONGODB_URI=mongodb+srv://suryapsgtech_db_user:ChummaZ4001@cluster0.owtcttv.f
2 #MONGODB_URI=mongodb+srv://sumitha:Testleaf321@cluster0.c514wjg.mongodb.net/
3 DB_NAME="db_testcases"
4 #DB_NAME="RagTest_cases"
5
6 COLLECTION_NAME="TestCases"
7 VECTOR_INDEX_NAME="vector_index_test_cases"
8 BM25_INDEX_NAME="bm25_search"
9
10 USER_STORIES_COLLECTION_NAME="UserStories"
11 USER_STORIES_VECTOR_INDEX_NAME="vector_index_user_stories"
12
13 #USER_STORIES_COLLECTION_NAME="user_stories"
14 #USER_STORIES_VECTOR_INDEX_NAME="vector_index"
15
16 # Mistral AI Configuration (for embeddings)
17 MISTRAL_API_KEY=wfrpypas0EIAjplvj2wyG5YrCtM8h2A
18 MISTRAL_EMBEDDING_MODEL="mistral-embed"
```

The terminal panel at the bottom shows the output of the 'npm run dev' command:

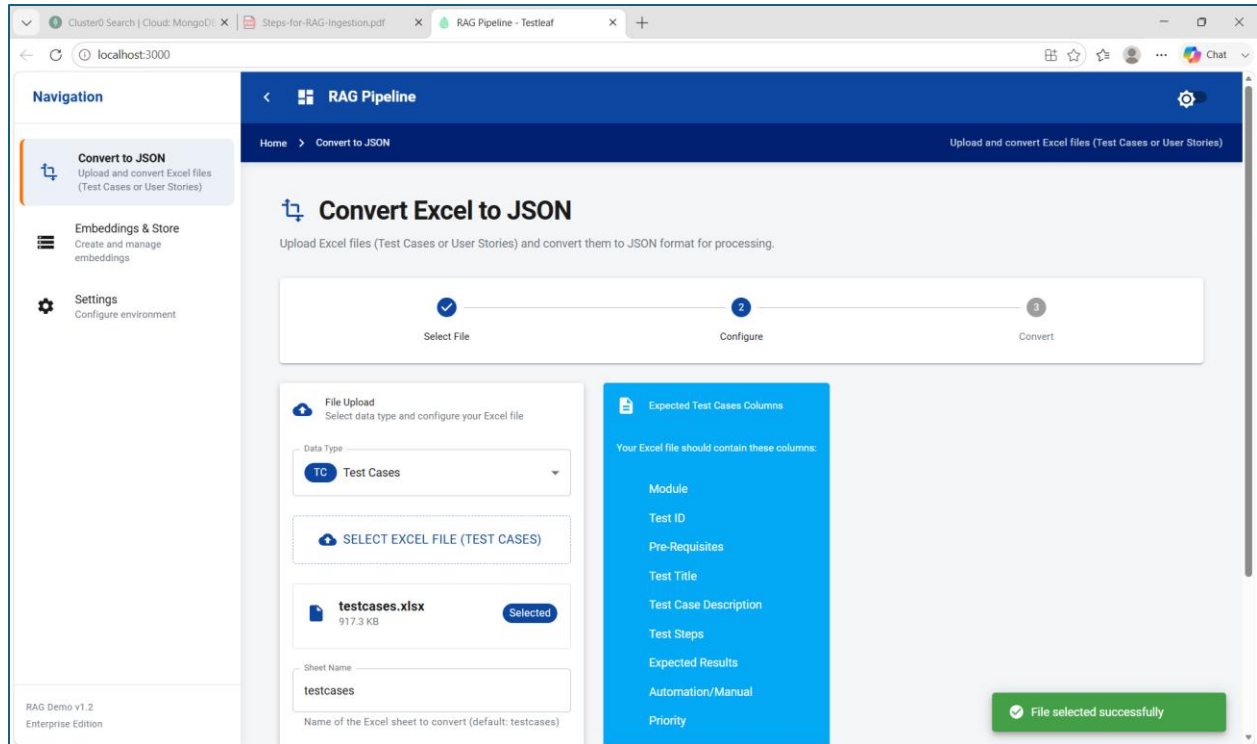
```
PS C:\GenAIForQA\Phase2\Week03\rag-mongo-demo-ui\rag-mongo-demo-ui> npm run dev
[1] (Use "node --trace-deprecation ..." to show where the warning was created)
[1] Browserslist: browsers data (caniuse-lite) is 9 months old. Please run:
[1]   npx update-browserslist-db@latest
[1]   Why you should do it regularly: https://github.com/browserslist/update-db#readme
[1] (node:15624) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterS
[1] etupMiddlewares' option is deprecated. Please use the 'setupMiddlewares' option.
[1] (node:15624) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBefor
[1] eSetupMiddlewares' option is deprecated. Please use the 'setupMiddlewares' option.
[1] Starting the development server...
[1]
[1] Compiled successfully!
[1]
[1] You can now view client in the browser.
[1]
[1] Local:      http://localhost:3000
[1] On Your Network:  http://192.168.1.137:3000
[1]
[1] Note that the development build is not optimized.
[1] To create a production build, use npm run build.
[1]
[1] webpack compiled successfully
```

The right sidebar shows the 'CHAT' panel with a 'SESSIONS' section and a 'Tip: Try the Plan agent to research and plan before implementing changes.' message.

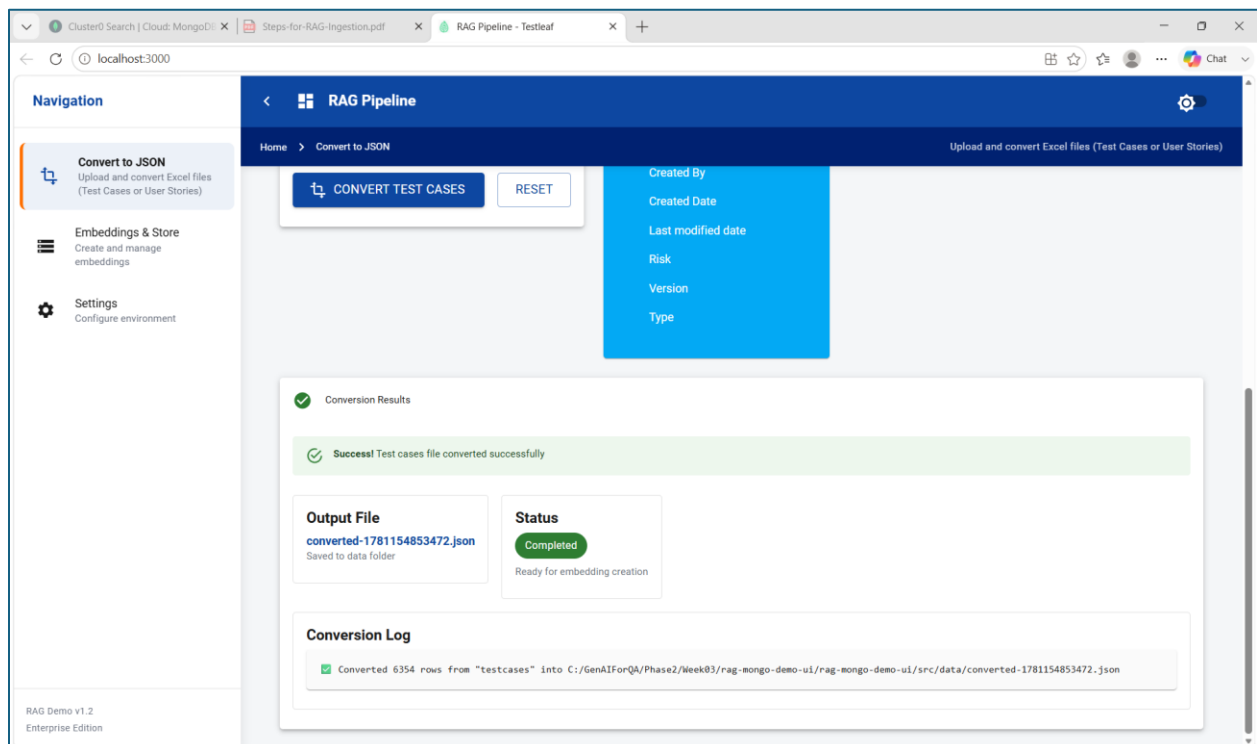
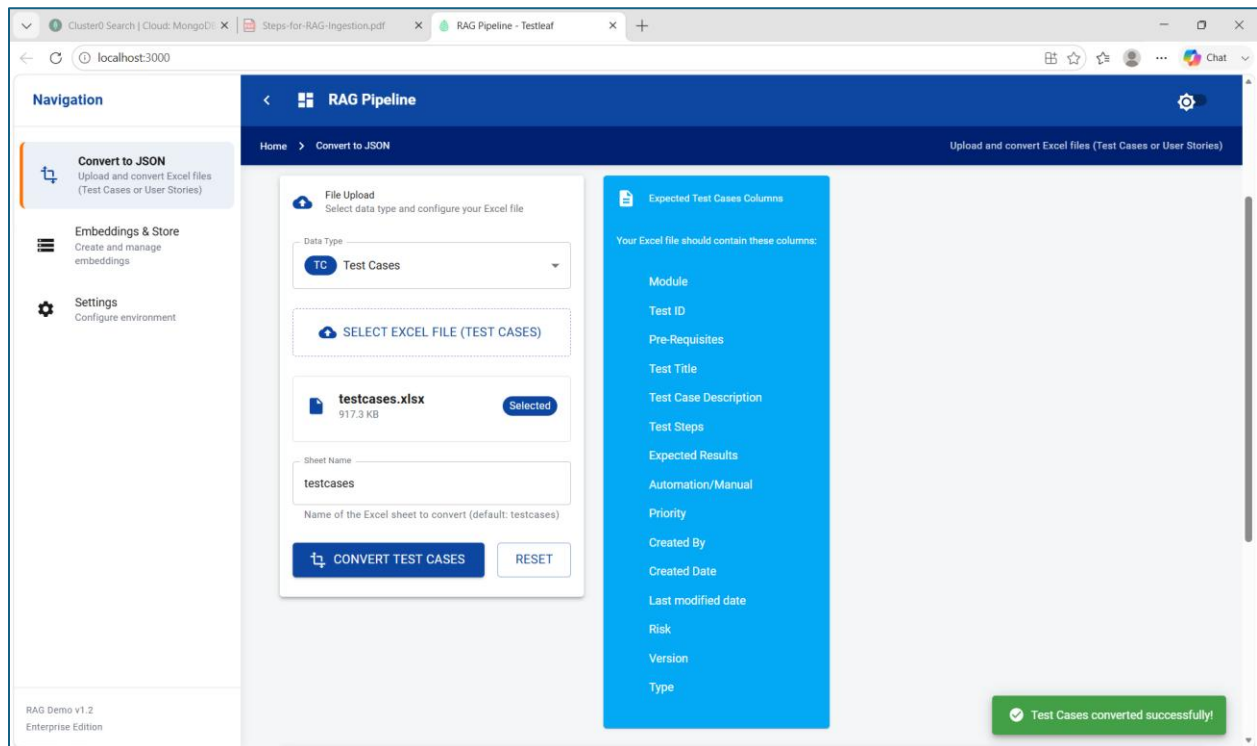
UI for RAG Ingestion



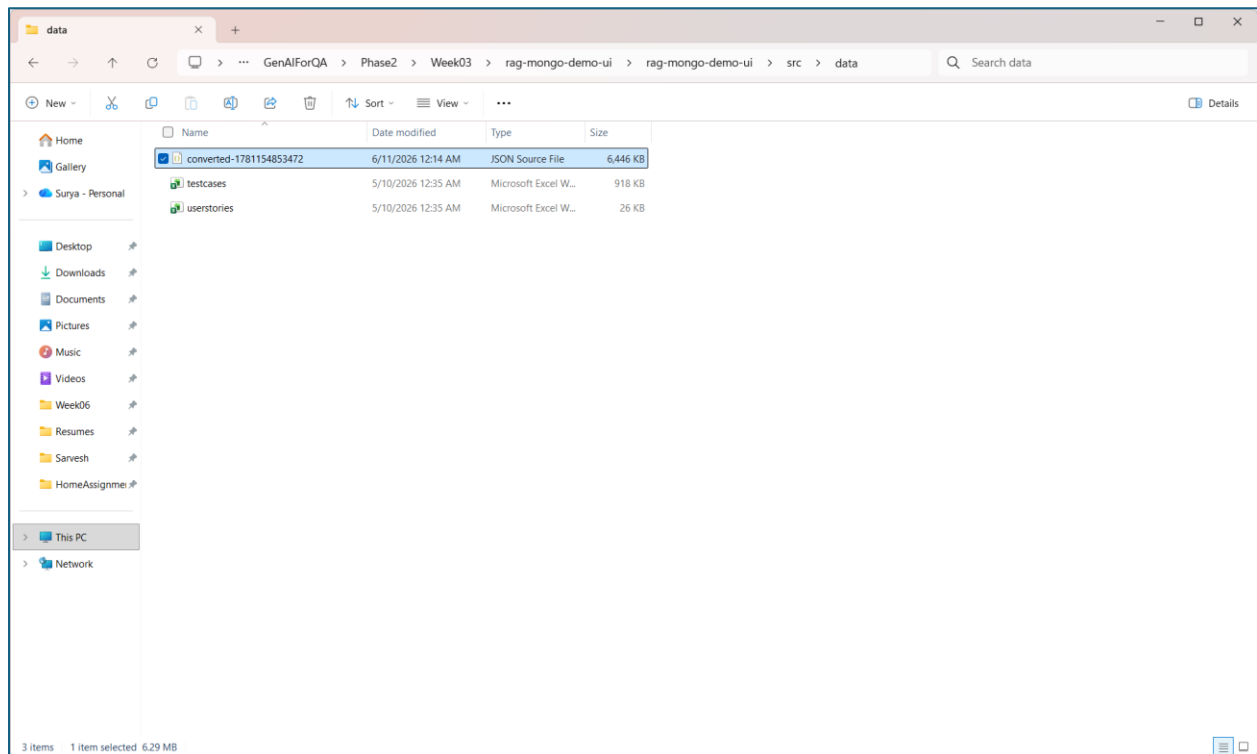
Step1: Select DataType – Testcases and select the excel file



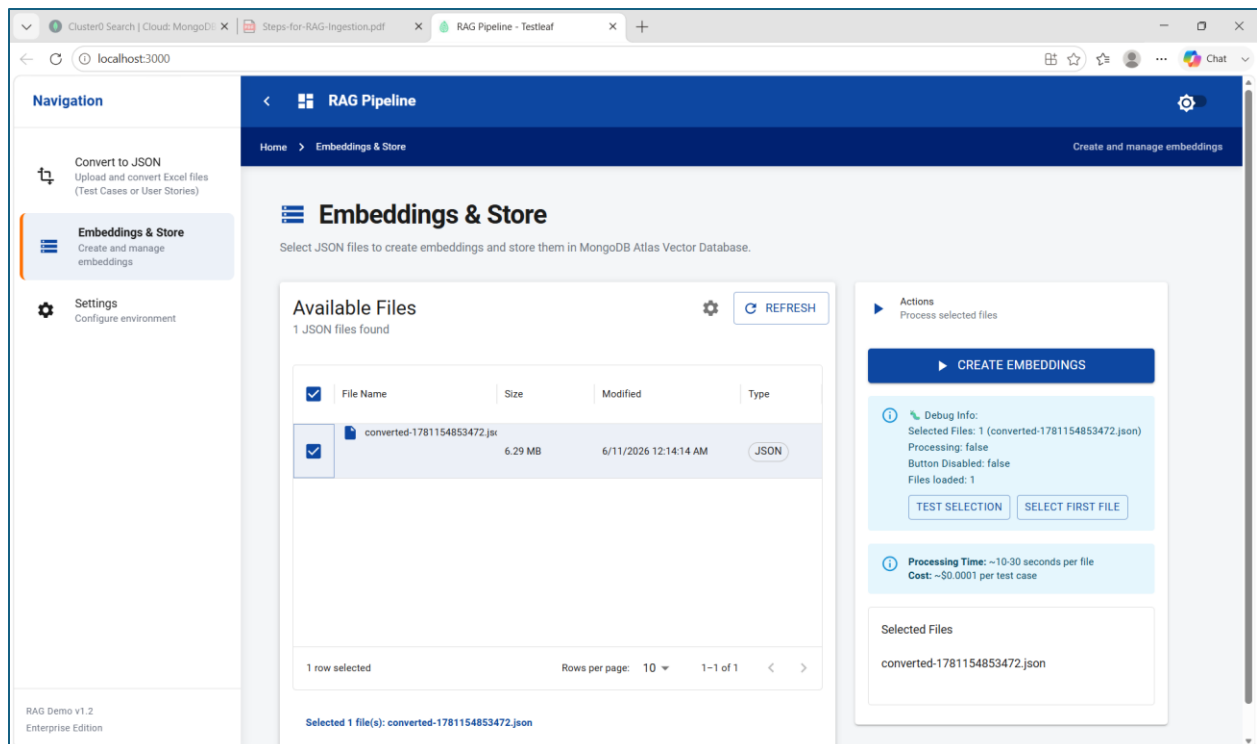
Step2: Click Convert Testcases button



Converted JSON in the path src\data



Step3: Select Embeddings and Store



Step4: Click on Create Embeddings

The screenshot shows the 'RAG Pipeline' web interface. The left sidebar contains navigation options: 'Convert to JSON', 'Embeddings & Store' (selected), and 'Settings'. The main area is titled 'Embeddings & Store' and includes a description: 'Select JSON files to create embeddings and store them in MongoDB Atlas Vector Database.' Below this, there's a section for 'Available Files' showing 1 JSON file found. A table lists the file 'converted-1781154853472.json' with details like size (6.29 MB) and modified date (6/11/2026 12:14:14 AM). To the right, there's a 'PROCESSING...' status bar and a 'Debug Info' section showing 'Selected Files: 1 (converted-1781154853472.json)' and 'Processing: true'. A green notification bar at the bottom states 'Batch processing started! Job ID: job-1781154977161-71t20lo4q'.

Batch process starts

The screenshot shows a VS Code editor with the 'user-stories-vector-index.json' file open. The file contains configuration for MongoDB, Mistral AI, and vector search. The terminal output shows the execution of 'npm run dev' and the start of batch processing for embeddings. The output includes details about the batch size (50 testcases), success status, cost, and tokens used.

```
PS C:\GenAIForQA\Phase2\Week03\rag-mongo-demo-ui> npm run dev
[0] [testcases] [Batch 5/128] Success! Cost: $0.000794 | Tokens: 7941
[0] [testcases] [Batch 6/128] Success! Cost: $0.000895 | Tokens: 8945
[0] [testcases] [Batch 4/128] Success! Cost: $0.000937 | Tokens: 9374
[0] [testcases] [Batch 7/128] Processing 50 testcases with Mistral AI...
[0] [testcases] [Batch 8/128] Processing 50 testcases with Mistral AI...
[0] [testcases] [Batch 9/128] Success! Cost: $0.000896 | Tokens: 8955
[0] [testcases] [Batch 7/128] Success! Cost: $0.000925 | Tokens: 9248
[0] [testcases] [Batch 8/128] Success! Cost: $0.000992 | Tokens: 9920
```

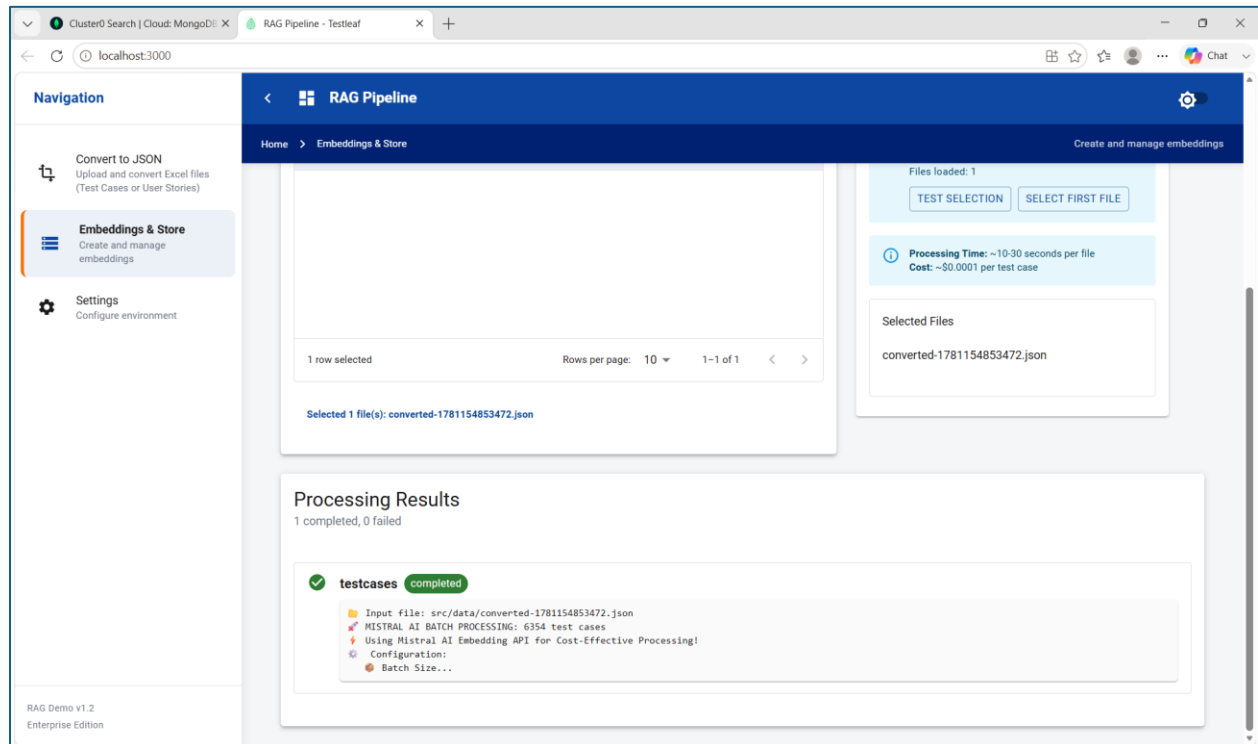
Successful Completion of batch process

The screenshot shows a Visual Studio Code editor with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project named 'RAG-MONGO-DEMO-UI' with various files and folders. The code editor displays a file named 'user-stories-vector-index.json' with the following content:

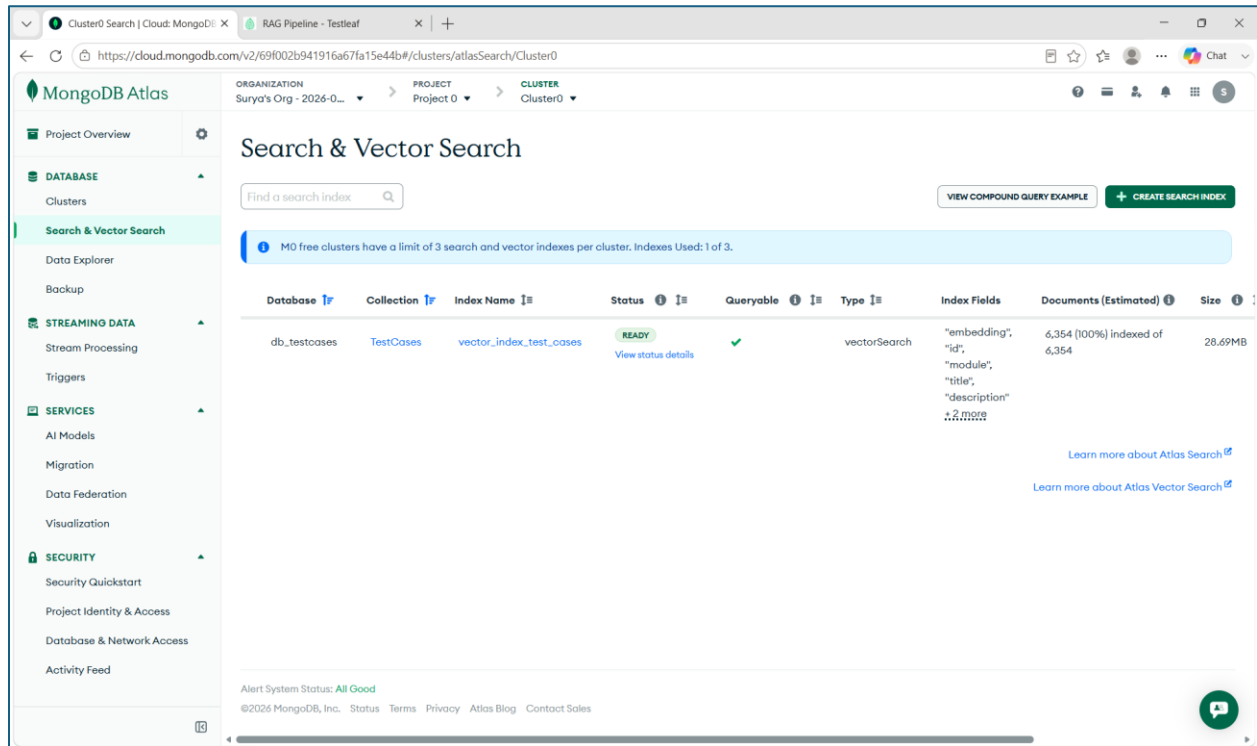
```
1 MONGODB_URI=mongodb+srv://suryapsgetch_db_user:chummaK4001@cluster0.owtcttv.mongodb.net/
2 #MONGODB_URI=mongodb+srv://sumitha:TestLeaf321@cluster0.c514wjq.mongodb.net/
3 DB_NAME="db_testcases"
4 #DB_NAME="RagTest_cases"
5
6 COLLECTION_NAME="TestCases"
7 VECTOR_INDEX_NAME="vector_index_test_cases"
8 BM25_INDEX_NAME="bm25_search"
9
10 USER_STORIES_COLLECTION_NAME="UserStories"
11 USER_STORIES_VECTOR_INDEX_NAME="vector_index_user_stories"
12
13 #USER_STORIES_COLLECTION_NAME="user_stories"
14 #USER_STORIES_VECTOR_INDEX_NAME="vector_index"
15
16 # Mistral AI Configuration (for embeddings)
17 MISTRAL_API_KEY=wfrpypa50EIAjplVj2wyG5YrCtW8h2A
18 MISTRAL_EMBEDDING_MODEL="mistral-embed"
```

The terminal at the bottom shows the output of the command 'npm run dev'. The output indicates that the batch processing is complete and provides the following statistics:

```
PS C:\GenAIForQA\Phase2\Week03\rag-mongo-demo-ui\rag-mongo-demo-ui> npm run dev
[0] [testcases]
[0] MISTRAL AI BATCH PROCESSING COMPLETE!
[0] Final Statistics:
[0]   Total Time: 4m 21s
[0]   Processing Rate: 24.3 testcases/second
[0]   Total Test Cases: 6354
[0]   Successfully Processed: 6354
[0]   Failed: 0
[0]   Success Rate: 100.0%
[0]   Total cost: $0.117741
[0] [testcases]   Total Tokens: 1,177,411
[0] [testcases]   Average Cost per Test Case: $0.0001853
[0] [testcases]   Average Tokens per Test Case: 185
[0] [testcases]   Cost Savings: ~90% cheaper than OpenAI embeddings
[0] [testcases]   Batch script completed successfully!
```

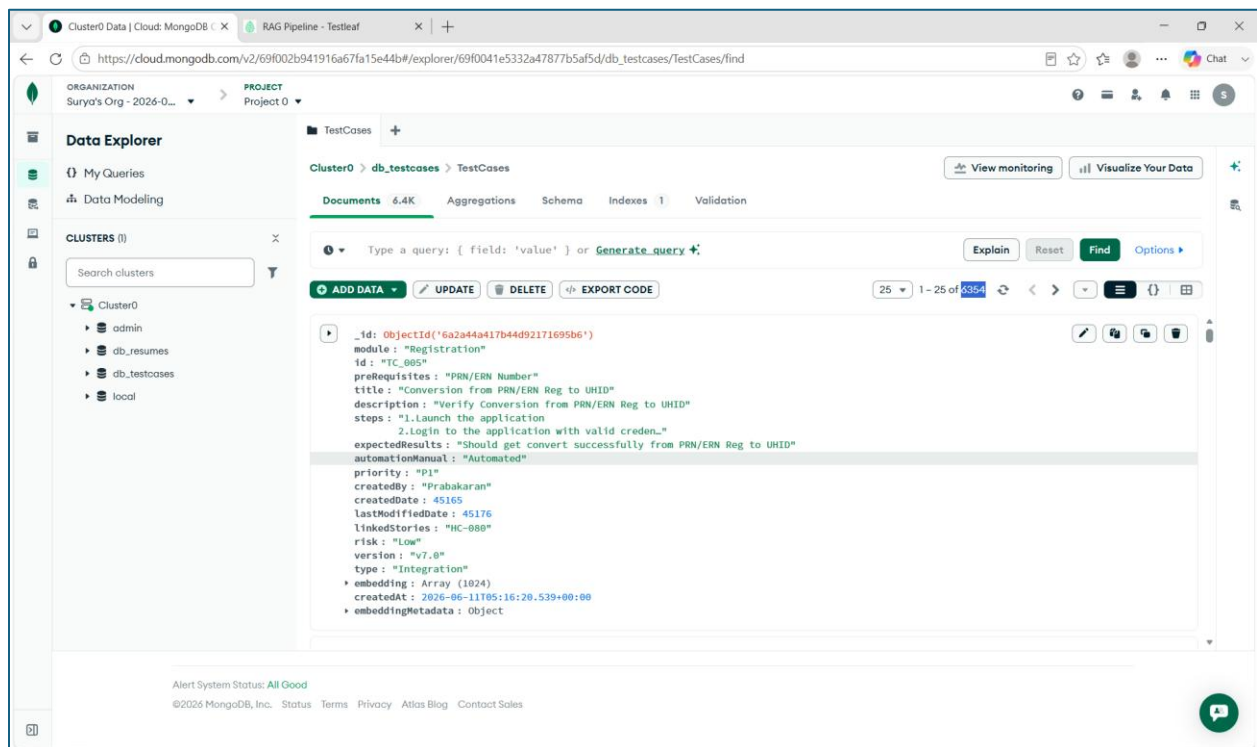


MongoDB Verification



The screenshot shows the MongoDB Atlas 'Search & Vector Search' interface. The left sidebar contains navigation links for Project Overview, DATABASE (Clusters), Search & Vector Search (selected), Data Explorer, Backup, STREAMING DATA (Stream Processing, Triggers), SERVICES (AI Models, Migration, Data Federation, Visualization), and SECURITY (Security Quickstart, Project Identity & Access, Database & Network Access, Activity Feed). The main content area has a search bar and a table of search indexes. A blue banner at the top states: 'M0 free clusters have a limit of 3 search and vector indexes per cluster. Indexes Used: 1 of 3.' The table lists one index: 'vector_index_test_cases' in the 'TestCases' collection of the 'db_testcases' database. It is a 'vectorSearch' type, 'READY' status, and 'Queryable'. The index fields are 'embedding', 'id', 'module', 'title', and 'description'. It contains 6,354 documents (100% indexed) and has a size of 28.69MB. Links for 'Learn more about Atlas Search' and 'Learn more about Atlas Vector Search' are provided. The footer shows 'Alert System Status: All Good' and copyright information for MongoDB, Inc. 2026.

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size
db_testcases	TestCases	vector_index_test_cases	READY	✓	vectorSearch	"embedding", "id", "module", "title", "description"	6,354 (100%) indexed of 6,354	28.69MB



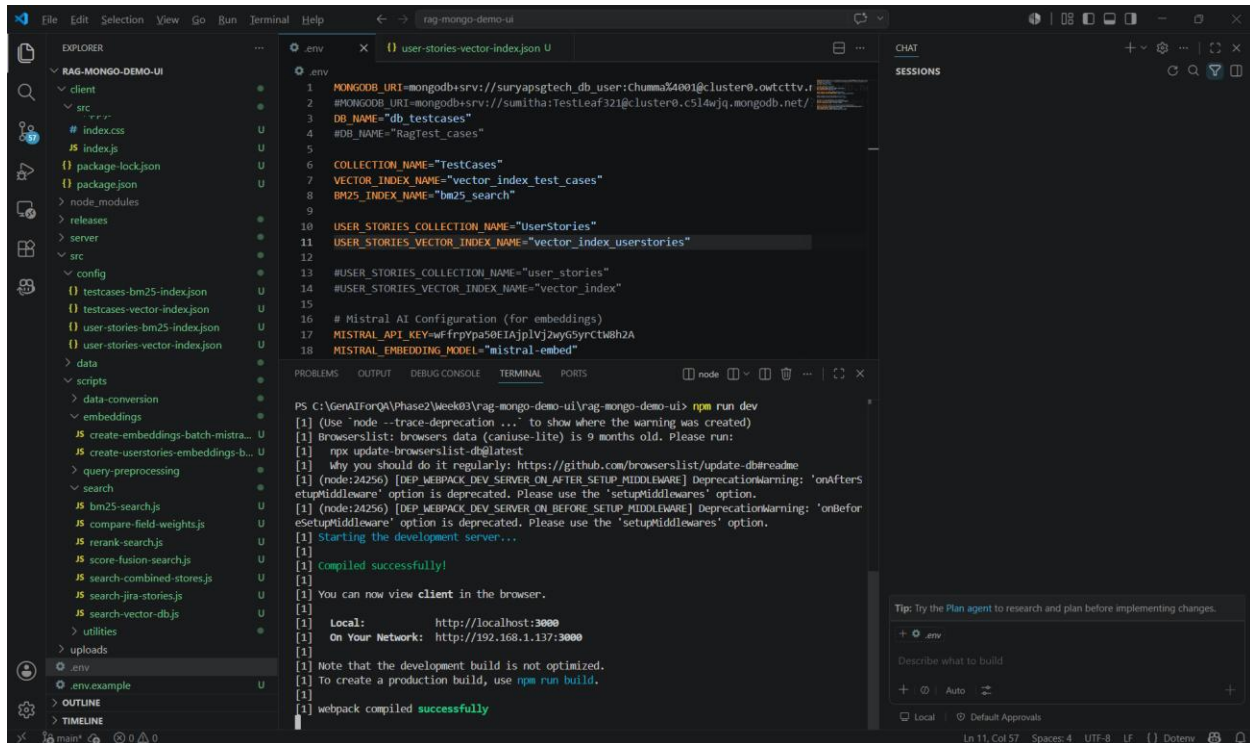
The screenshot shows the MongoDB Atlas 'Data Explorer' interface. The left sidebar has links for My Queries, Data Modeling, and CLUSTERS (Cluster0 selected). The main content area shows the 'TestCases' collection with 6.4K documents. A query editor is active with a JSON query:

```
{ "_id": ObjectId('6a2a44a17b44d92171695b6')
  "module": "Registration"
  "id": "TC_005"
  "preRequisites": "PRN/ERN Number"
  "title": "Conversion from PRN/ERN Reg to UHID"
  "description": "Verify Conversion from PRN/ERN Reg to UHID"
  "steps": "1.Launch the application
            2.Login to the application with valid creden..."
  "expectedResults": "Should get convert successfully from PRN/ERN Reg to UHID"
  "automationManual": "Automated"
  "priority": "P1"
  "createdBy": "Prabakaran"
  "createdDate": 45165
  "lastModifiedDate": 45176
  "linkedStories": "HC-000"
  "risk": "Low"
  "version": "v7.0"
  "type": "Integration"
  "embedding": Array (1024)
  "createdAt": 2026-06-11T05:16:20.539+00:00
  "embeddingMetadata": Object
```

. The interface includes buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'. The footer shows 'Alert System Status: All Good' and copyright information for MongoDB, Inc. 2026.

RAG Ingestion – Frontend for Userstories

Starting the frontend server for userstories ingestion



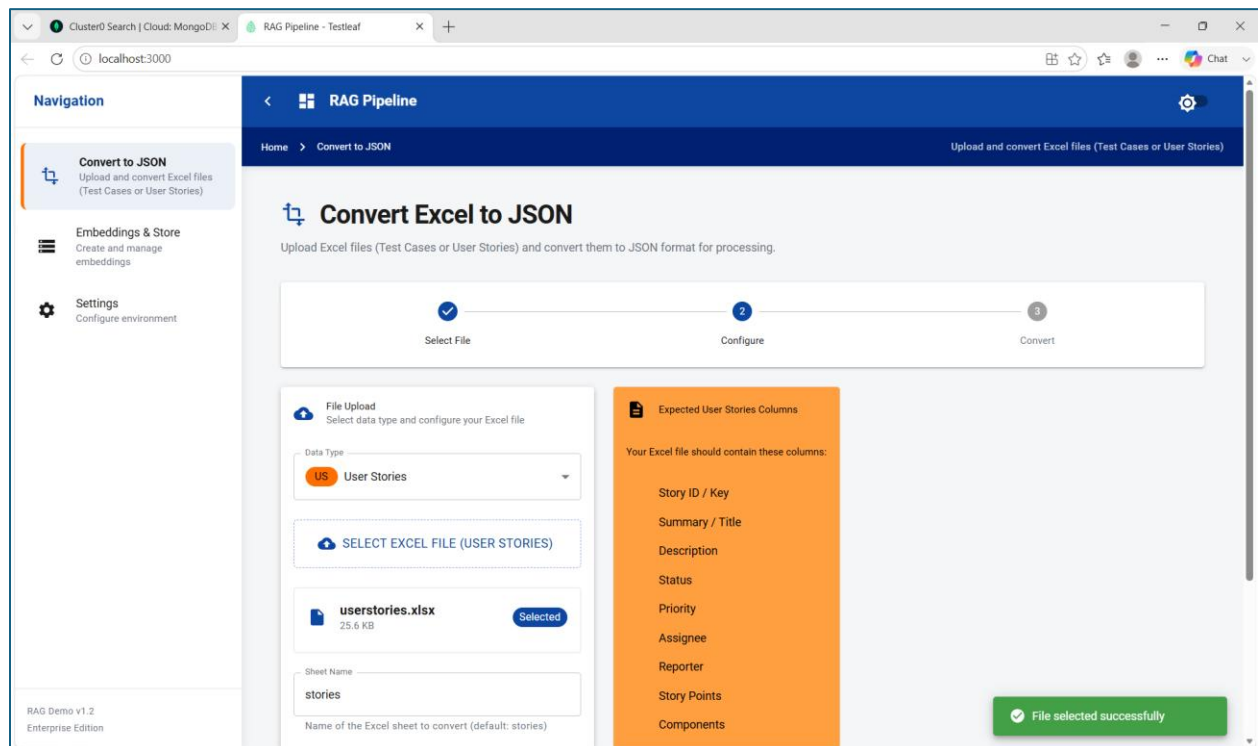
The screenshot shows the VS Code interface for the 'rag-mongo-demo-ui' project. The Explorer pane on the left displays the project structure, including 'client', 'server', 'config', 'data', 'scripts', 'embeddings', and 'utils'. The main editor shows the 'user-stories-vector-index.json' file with the following content:

```
.env
1 MONGODB_URI=mongodb+srv://suryapsgtech_db_user:chumma34001@cluster0.owtcttv.mongodb.net/
2 #MONGODB_URI=mongodb+srv://sumitha:Testleaf321@cluster0.c514wjq.mongodb.net/
3 DB_NAME="db_testcases"
4 #DB_NAME="RagTest_cases"
5
6 COLLECTION_NAME="TestCases"
7 VECTOR_INDEX_NAME="vector_index_test_cases"
8 BM25_INDEX_NAME="bm25_search"
9
10 USER_STORIES_COLLECTION_NAME="UserStories"
11 USER_STORIES_VECTOR_INDEX_NAME="vector_index_userstories"
12
13 #USER_STORIES_COLLECTION_NAME="user_stories"
14 #USER_STORIES_VECTOR_INDEX_NAME="vector_index"
15
16 # Mistral AI Configuration (for embeddings)
17 MISTRAL_API_KEY=hfryps0EIAjplVj2yG5yrcT8h2A
18 MISTRAL_EMBEDDING_MODEL="mistral-embed"
```

The terminal output shows the command 'npm run dev' being executed, resulting in a successful compilation and the development server starting on localhost:3000.

```
PS C:\GenAIForQA\Phase2\week03\rag-mongo-demo-ui\rag-mongo-demo-ui> npm run dev
[1] (Use "node --trace-deprecation ..." to show where the warning was created)
[1] Browserslist: browsers data (caniuse-lite) is 9 months old. Please run:
[1]   npm update browserslist
[1] Why you should do it regularly: https://github.com/browserslist/update-db#readme
[1] (node:24256) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
[1] (node:24256) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBeforeSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
[1] Starting the development server...
[1] compiled successfully!
[1] You can now view client in the browser.
[1] Local: http://localhost:3000
[1] On Your Network: http://192.168.1.137:3000
[1] Note that the development build is not optimized.
[1] To create a production build, use npm run build.
[1] webpack compiled successfully
```

Step1: Select Datatype – UserStories and then select the user stories excel file



Step2: Click on Convert User Stories

Navigation

- Convert to JSON**
Upload and convert Excel files (Test Cases or User Stories)
- Embeddings & Store
Create and manage embeddings
- Settings
Configure environment

RAG Pipeline

Home > Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Progress: Select File (✓) | Configure (✓) | Convert (✓)

File Upload
Select data type and configure your Excel file

Data Type: **US** User Stories

SELECT EXCEL FILE (USER STORIES)

userstories.xlsx
25.6 KB
Selected

Sheet Name: **stories**
Name of the Excel sheet to convert (default: stories)

CONVERT USER STORIES **RESET**

Expected User Stories Columns

Your Excel file should contain these columns:

- Story ID / Key
- Summary / Title
- Description
- Status
- Priority
- Assignee
- Reporter
- Story Points
- Components
- Labels
- Sprint
- Epic
- Acceptance Criteria
- Created Date

User Stories converted successfully!

Navigation

- Convert to JSON**
Upload and convert Excel files (Test Cases or User Stories)
- Embeddings & Store
Create and manage embeddings
- Settings
Configure environment

RAG Pipeline

Home > Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Conversion Results

Success! User stories file converted successfully

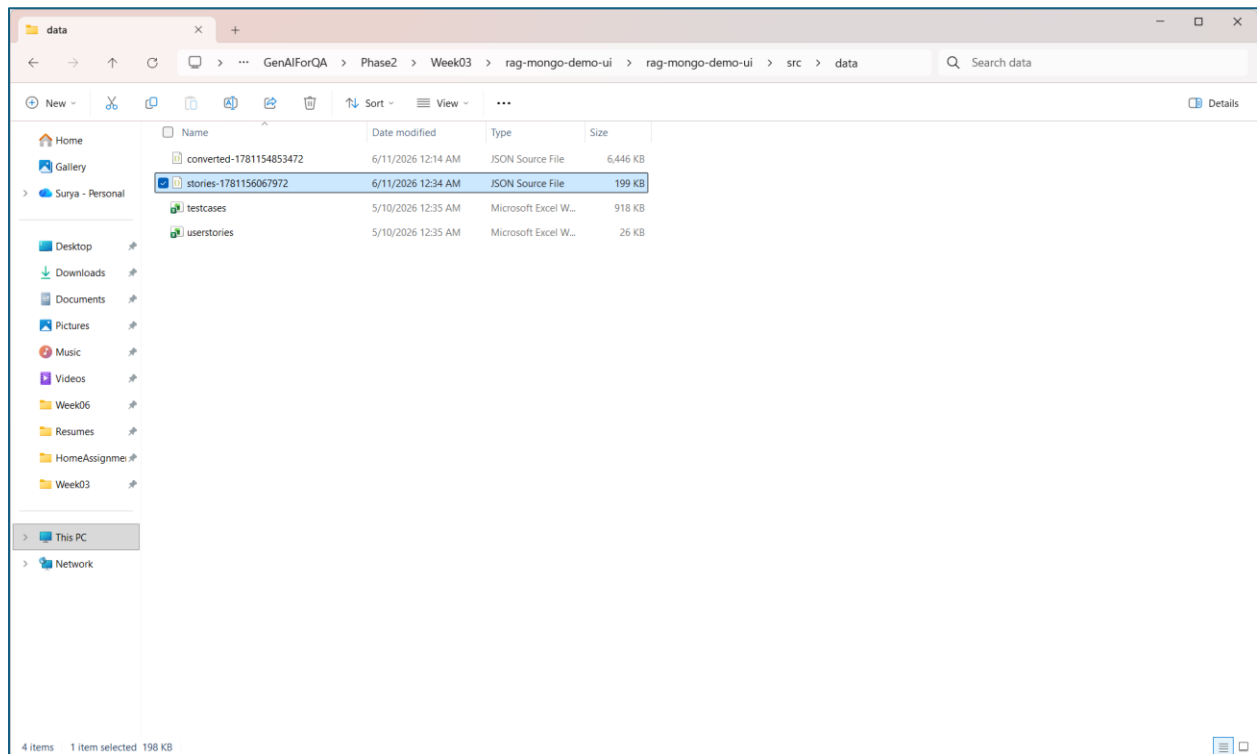
Output File
stories-1781156067972.json
Saved to data folder

Status
Completed
Ready for embedding creation

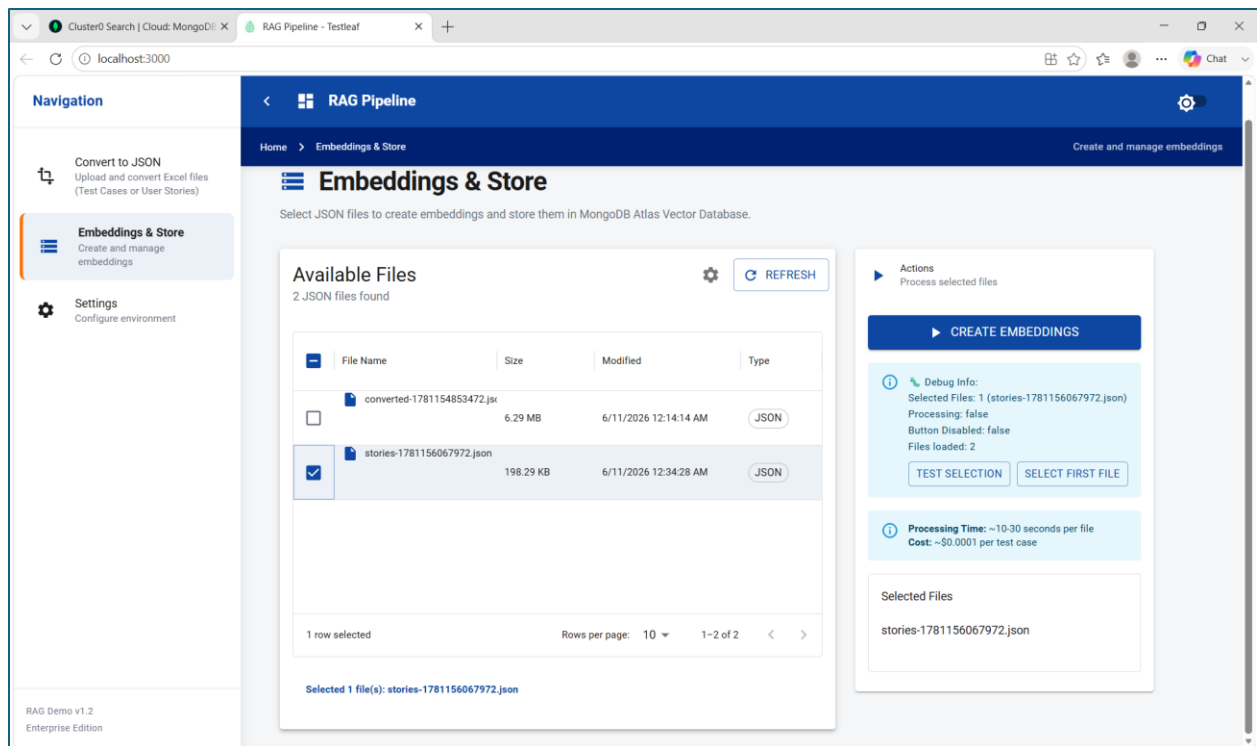
Conversion Log

- Converting 190 rows from sheet "stories"...
- Converted 190 user stories from "stories" into C:/GenAIForQA/Phase2/Week03/rag-mongo-demo-ui/rag-mongo-demo-ui/src/data/stories-1781156067972.json
- Filtered out: 0 empty rows

Converted json file is available in \src\data



Step3: Select Embedding and Store option and Select the converted stories json file



Step4: Select Create Embeddings option

The screenshot shows the 'RAG Pipeline' interface with the 'Embeddings & Store' section active. The 'Available Files' table lists two JSON files:

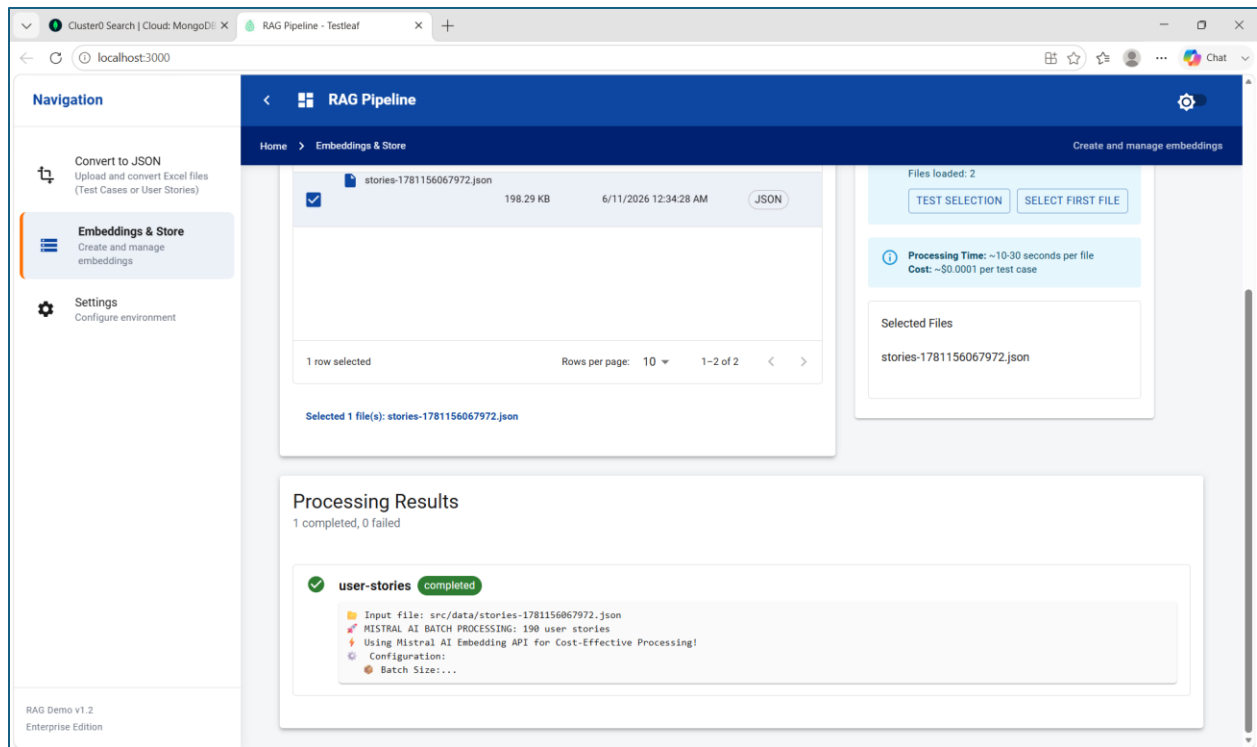
File Name	Size	Modified	Type
converted-1781154853472.json	6.29 MB	6/11/2026 12:14:14 AM	JSON
stories-1781156067972.json	198.29 KB	6/11/2026 12:34:28 AM	JSON

The 'stories-1781156067972.json' file is selected. The 'Actions' panel on the right shows 'PROCESSING...' and 'Batch processing started!' messages. The 'Debug Info' section shows 'Selected Files: 1 (stories-1781156067972.json)' and 'Processing: true'. The 'Processing Time' is '~10-30 seconds per file' and the 'Cost' is '~\$0.0001 per test case'. The 'Selected Files' section shows 'stories-1781156067972.json'. A green notification bar at the bottom says 'Batch processing started! Job ID: job-1781156219740-mkwgwgrf95'. A blue button at the bottom right says 'Starting batch processing for 1 files...'.

Batch Processing starts and completes successfully

The screenshot shows a VS Code terminal window with the output of a batch processing script. The script successfully processes 190 user stories, with a total time of 10s, a processing rate of 19.0 user stories/second, and a total cost of \$0.003068. The output includes the following information:

```
PS C:\GenAIForQA\Phase2\Week03\rag-mongo-demo-ui\rag-mongo-demo-ui> npm run dev
[0] Progress: 190/190 (100.0%) | Rate: 24.0/sec | ETA: 0s | Cost: $0.003068
[0] [user-stories]
[0] # MISTRAL AI BATCH PROCESSING COMPLETE!
[0] # Final Statistics:
[0] [user-stories] Total Time: 10s
[0] Processing Rate: 19.0 user stories/second
[0] Total User Stories: 190
[0] Successfully Processed: 190
[0] Failed: 0
[0] Success Rate: 100.0%
[0] Total cost: $0.003068
[0] [user-stories] Total Tokens: 30,678
[0] [user-stories] Average Cost per User Story: $0.00001615
[0] Average Tokens per User Story: 161
[0] Cost Savings: ~98% cheaper than OpenAI embeddings
[0] Batch script completed successfully!
```



MongoDB Verification

